DPP	DEPARTMENTAL POLICY AND PROCEDURE		
Title	**CARE AND MANAGEMENT OF NEWBORN ON MECHANICAL VENTILATOR**		
Department	**Neonatology Department**		
Applies to	**All NICU Medical and Nursing Staff, Respiratory Therapist**		
Policy Code and Version Number	**DPP-NICU-006 (v2)**		
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19 / 3 / 2024	7 / 4 / 2024	TRIENNIAL	18 / 3 / 2027

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1.0 PURPOSE

1.1. To provide a process for the respiratory management of neonatal patients in Neonatal Intensive Care Unit (NICU).

1.2. To establish and ensure correct ventilator parameters for newborn who require mechanical ventilation.

1.3. To provide procedural guidelines on how to give nursing care to newborn on mechanical ventilator.

2.0 DEFINITION/ABBREVIATION

2.1. **Practice outcome:** To provide appropriate ventilation to minimize ventilator induced lung injury (barotrauma, volutrauma, atelectotrauma, oxygen toxicity) and optimize oxygenation to prevent chronic lung disease.

2.2. **Mechanical Ventilation** is the medical term for artificial ventilation where machine is used to assist or replace spontaneous breathing of the newborn.

2.3. **Chest Physiotherapy** is the rhythmically striking of the thorax over the lung segment to aid in the indirect removal of mucus from the breathing passages of the newborn.

2.4. **PEEP** – Positive End Expiratory Pressure

2.5. **PIP** –Peak Inspiratory Pressure

2.6. **FiO2** – Fraction of Inspired Oxygen Concentration

2.7. **Pa02** - Partial pressure of oxygen

2.8. **HFV**- High Frequency Ventilation

2.9. **I: E** - Inspiratory /Expiratory ratio

2.10. **MAP**- Mean Airway Pressure

2.11. **PPE**- Personal Protective Equipment

2.12. **NICU**-Neonatal Intensive Care Unit

3.0 RESPONSIBILITY

3.1. **Physician** (Resident, Specialist, Consultant) are responsible to provide a written order of the ventilator parameters to be set in the medical file.

3.2. Nurse (Head Nurse, Charge Nurse, Staff Nurse) are responsible to give nursing care to the newborn on mechanical ventilation.

3.3. The nurse will document the procedure done and the ventilator settings in the medical file.

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3.4. **Respiratory Therapist** monitors the neonate carefully to optimize safety and comfort, and to ensure that the duration of mechanical ventilation is minimized as much as possible.

4.0 POLICY

4.1. To determine ventilation strategy, consider:

4.1.1. Pathophysiology of the respiratory disease process

4.1.2. Clinical assessment of the patient

4.1.3. The limitations of the ventilator chosen

4.1.4. Patient ventilator interaction

4.1.5. Oxygen saturation

4.1.6. Ventilation-based respiratory mechanics

4.1.7. Blood gases

4.2. Ventilation plans, documented in orders and progress notes include:

4.2.1. Mode of ventilation

4.2.2. Initial ventilator settings

4.2.3. Parameters for potential weaning or escalation if necessary

4.2.4. Timing of blood gas sampling

4.3. All newborn requiring mechanical ventilation must be admitted to Neonatal Intensive Care Unit.

4.4. Setting up of ventilator parameters must be according to the physician's written order.

4.5. All physician and nurses caring for the newborn on mechanical ventilation must be certified in Basic Life Support and Neonatal Resuscitation Program.

4.6. Nurse: Ventilated newborn ratio must be 1:1or to a maximum of 1:2 according to staffing plan.

4.7. Mechanical Ventilator parameters:

4.7.1. **Respiratory Distress Syndrome**:

4.7.1.1. Fi02: 40-100 %, depending on the clinical situation

4.7.1.2. Rate: 30 - 40 per/minute

4.7.1.3. PEEP: 5-6 cm H20 or 4 cm H20 if Fi02 is above 90%

4.7.1.4. PIP: Will be determined by adequate chest wall movement.

4.7.1.4.1. 16 - 28 cm H20- for newborn weighing less than 1500 grams

4.7.1.4.2. 20 -30 cm H20- for newborn weighing greater than 1500 grams

4.7.1.5. Inspiratory Time: 0.3 - 0.5 seconds

4.7.1.6. If the Pa02 or the oxygen saturation is below accepted standards, the Fi02 can be raised to a maximum of 100%.

4.7.1.7. If the Pa02 or 02 saturations is still inadequate, the mean airway pressure can be raised by increasing either the PIP, PEEP, inspiratory time or the rate, leaving the inspiratory time constant.

4.7.1.8. If the PaC02 is elevated, the rate or peak inspiratory pressure can be raised.

4.7.1.9. When lowering the respiratory rate without a concomitant decrease in I: E ratio, the inspiratory time can become quite prolonged.

4.7.1.9.1. The total Inspiratory time should not exceed 0.6 seconds.

4.7.1.9.1.1. When increasing the respiratory rate above 60/minute, the I: E ratio should be 1:1

4.7.2. **Other Respiratory Conditions**:

4.7.2.1. The peak pressure used is a reflection of the anticipated compliance of the lung

4.7.2.2. Subsequent changes in settings will be determined by arterial blood gases and the newborn's clinical course.

4.8. When setting up the initial parameters for MV, the physician must indicate the following in the file:

4.8.1. Mode (IMV or conventional sigh breaths when using HFV)

4.8.2. Rate (breaths per minute)

4.8.3. Fi02

4.8.4. Inspiratory time (seconds) or I: E ratio

4.8.5. Peak Inspiratory Pressure

4.8.6. PEEP

4.8.6.1. High Frequency Ventilation (HFV)

4.8.6.1.1. Frequency

4.8.6.1.2. Amplitude or power

4.8.6.1.3. PEEP or MAP

4.8.6.2. Synchronized Intermittent Mandatory Ventilation (SIMV)

4.8.6.2.1. Rate (use up to 40 breaths per minutes when on servo 300, up to 60 om star synch)

4.8.6.2.2. Pressure control: set a peak pressure based on adequate chest wall movement

4.8.6.2.3. Pressure support: number of cm H_2O pressure above the PEEP

4.8.6.2.4. Usually starts at a PS: (PIP-PEEP)/2, minimal PS: 4-6 cm

4.8.6.2.5. Volume control: set a tidal volume usually 5-7 ml/kg for premature infants and T-10 ml/kg for term infants.

4.9. Arterial Blood Gases must be done within 30 minutes of initiating mechanical ventilation or making a parameter change or as per physician order.

4.10. Newborn on mechanical ventilation must have a documentation of accurate daily weight and record of precise Intake and Output.

4.11. Disinfection of the mechanical ventilator must be according to Infection Prevention and control policy.

4.12. Assess patient and monitor pulmonary graphics for:

4.12.1. Chest over-distention / hyperinflation (q1h).

4.12.2. Blood pressure changes resulting from change in ventilation. (q1h) if arterial line present.

4.12.3. Pain and agitation (at least q4h).

4.13. Immediately following intubation, if medications to induce paralysis (RSI) have been used either for the intubation or for other indications, set Vg to 4.5 ml/kg and set the rate to 50 breaths per minute (bpm) to prevent hypercarbia. If infant remains paralyzed, then do an arterial blood gas as soon as possible (within ~ 20 mins) following the intubation to assess mechanical ventilation needs.

4.14. If the ETT leak is > 40% investigate the cause of the leak and if necessary, consider reintubation with a larger tube unless there is a clinical contraindication to the procedure.

4.15. Routine Blood Gas Parameters Targets: (arterial and capillary except where indicated)

4.15.1. PO2: 50-70mmHg (arterial only)

4.15.2. PCO2: 45- 55 mmHg (may be higher as long as pH is >7.2)

4.15.3. pH: > 7.2

4.15.4. HCO3: > 17.5 mmol/l

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4.15.5. BD: no lower than -6

4.15.6. BE: no higher than + 6

4.15.7. Lactate: < 2

4.15.8. Deviations from these should be made for cardiac patients and those with chronic lung disease.

4.15.9. Parameters will be set according to the pathophysiology and documented in the clinical record.

4.16. Determine frequency and timing of blood gas sampling with RT, physician/NNP/Clinical Assistant and bedside nurse in consultation.

4.17. Any time pCO2 is greater than 70 or less than 35 mmHg with a resulting pH <7.20, draw a blood gas no more than 30 minutes' post parameter change. In some cases, such as with BPD, pCo2 >70 and might be acceptable. Others with HIE, Co2 <35 as part of washing Co2 to compensate the metabolic acidosis and these are acceptable for such cases.

4.18. Change patient's position q4h or as tolerated. Include prone as tolerated.

4.19. Consider extubation according to the process and criteria

5.0 PROCEDURE (S)

5.1. The physician will assess the newborn requiring mechanical ventilation.

5.2. The nurse will verify the written order for mechanical ventilation in the medical file

5.3. Identify the patient correctly and perform hand hygiene according to policy

5.4. Prepare the mechanical ventilator and set up the parameters according to physician's written order.

5.5. Assist the physician to intubate the newborn according to policy if not yet done.

5.6. Attach the newborn to the ventilator and ensure that all connections are secured.

5.7. Set cardiorespiratory and ventilator alarms audible and according to newborn's baseline.

5.8. Provide the necessary supportive nursing care to the ventilated newborn.

5.8.1. Provide intensive monitoring of vital signs and adequate ventilation.

5.8.2. Change position to supine, right and left lateral or prone if feasible, every 2 - 4 hourly.

5.8.3. Provide effective chest physiotherapy as indicated.

5.8.4. Assess breath sounds and the need for suctioning every 2 hours or as directed by physician.

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5.9. Assess function of the mechanical ventilator at least every 6 hours or as necessary.

5.10. Change ventilator circuits every 72 hours or earlier if indicated

5.11. Provide accurate documentation of the procedures done to the newborn including:

5.11.1. Initial ventilator parameters set or any changes made in the settings

5.11.2. Date and time ventilator parameters were set up or changed

5.11.3. Any procedure done to newborn (e.g. Suctioning, physiotherapy, etc.)

5.11.4. Any changes in vital signs and signs of deterioration

5.11.5. Notification of physician about abnormal ABG/laboratory reports

5.12. **ASSIST CONTROL WITH VOLUME GUARANTEE (AC +Vg)**

5.12.1. **Selection Criteria**: Assist Control with Volume Guarantee (AC/Vg) is used for patients with moderate to severe lung disease.

5.12.2. **Management Guidelines**

5.12.2.1. Rate:

5.12.2.1.1. Infant breathing spontaneously starts at 40 breaths per minute (bpm).

5.12.2.1.2. Infants coasting with ventilator and severely depressed infant start at 50 bpm.

5.12.2.2. Inspiratory Time (Ti): 0.25 - 0.40 seconds

5.12.2.2.1. Take into consideration the flow waveform to identify best inspiratory time i.e. inspiration comes to zero flow with minimal pause to allow for best synchronization between ventilator and patient.

5.12.2.3. Flow: 6- 10 lpm. (Set to lower range for smaller patients and upper flow range for larger patients).

5.12.2.4. Positive End Expiratory Pressure (PEEP): +5 to + 7 cmH20 depending on pathology of the lung disease and cardiovascular profile of the patient.

5.12.2.5. Volume Guarantee (Vg): set 3.5 – 5 ml /kg. If Vg of 5.0ml/kg required, consider a chest x-ray to determine:

5.12.2.5.1. Position of endotracheal tube and lung inflation.

5.12.2.5.2. Whether using high frequency ventilation would improve ventilation and reduce ventilator-induced lung injury.

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5.13. PRESSURE SUPPORT (PC PSV + Vg)

5.13.1. **Selection Criteria:** Pressure Support Ventilation with Volume Guarantee (PSV + Vg) for patients with an intact respiratory drive who are recovering or who have mild or no lung disease. And to assess readiness for extubation after AC/VG.

5.13.2. **Management Guidelines**

5.13.2.1. Review the MAP after switching to PSV and if necessary increase the Positive End Expiratory Pressure (PEEP) to maintain MAP and stabilize lung volume. (Patient may become tachypneic if MAP is inadequate).

5.13.2.2. Set Ti 50% longer than the patient's measured spontaneous inspiratory time.

5.13.2.3. Set the backup rate to 10 - 20 bpm.

5.13.2.4. Initiate ventilation with volume guarantee (Vg) set at 4 ml/kg. Adjust within range 3 – 5 ml/ kg based on arterial blood gases and monitoring of pulmonary graphics.

5.13.2.5. For any patient under 40 weeks' gestation who require tidal volumes > 5ml/kg determine risk for volutrauma with data found in the ventilator menu. Consider alternate ventilation strategies (i.e. AC/VG or high frequency ventilation) or changing PEEP.

5.13.2.6. For small premature infants (approximately<1500 grams) use PSV+Vg for no more than 2 hours to evaluate respiratory drive and determine readiness for extubation.

5.14. NASAL CONTINUOUS POSITIVE AIRWAY PRESSURE (NCPAP)

5.14.1. **Selection Criteria**

5.14.1.1. NCPAP modes for patients with an intact respiratory drive who are recovering, who have mild, or no lung disease, or who are weaning from mechanical ventilation.

5.14.1.2. Trigger Pressure Assist (TrPA) or SNIPPV for very small infants or as a transition from other ventilation modes after extubation.

5.14.2. **Management Guidelines**

5.14.2.1. After extubation set PEEP at + 6 (or higher according to the infant's condition) cmH2O. If using TrPA set pressure assist at 9/+6 mmH2O and Ti at 0.3-0.4 seconds, rate 20 -30 per minute.

5.14.2.2. Maintain the CPAP during all procedures including routine hygiene, weighing, changing CPAP equipment etc. Two people are often required in order to ensure that FRC is not lost.

5.14.2.3. Perform hourly assessments:

5.14.2.3.1. Check the hat and prongs/mask ensuring that the nose and ears are in normal position and eyes are clearly visible.

5.14.2.3.2. Ensure generator is stable and secure.

5.14.2.3.3. Assess infant comfort.

5.14.2.3.4. Ensure delivery of the required level of NCPAP.

5.14.2.3.5. Assess work of breathing.

5.14.2.3.6. Record set and measured PEEP and FIO2.

5.14.2.3.7. Asses skin integrity.

5.14.2.4. Select optimum prong or mask size. Alternate prongs & mask q 3-6 hours. If prongs do not" fit"-alternate between 2 sizes of mask.

5.14.2.5. Use pacifiers and appropriate sized hats with chin straps as necessary to decrease air leak from the infant's open mouth.

5.14.2.6. Routine suctioning is not required. Assess patency of nares q3-4h. Clean prongs or mask with alcohol q6-8h. Allow to air dry and do not store in an airtight container.

5.14.2.7. Place indwelling orogastric tube and open to air (except for approximately 30-40 minutes' post-feeds) Use an 8Fr for babies >1000 gm weight and 6 Fr for those <1000 gm. Consider Aspirating the feeding tube to maintain patency and decompress the stomach q3-4h.

5.14.2.8. Place positioning devices such as a firm pad under the infant's chest to maintain optimum airway position.

5.14.2.9. Assess nares for skin breakdown q6-8h and more often if breakdown noted. Cover reddened areas with hydrocolloid dressing materials until skin is intact. Alternate between prongs and mask.

5.14.2.10. If infant < 28 weeks' gestation:

5.14.2.10.1. Start with mask.

5.14.2.10.2. Do not routinely alternate from mask to prongs for 24-48 hours after birth to prevent loss of FRC. Select whichever provides the best seal for that neonate.

5.14.2.10.3. Maintain ordered pressure (CPAP level) at all times during care and sleep.

5.15. HIGH FREQUENCY VENTILATION (HFV)

5.15.1. Selection Criteria

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5.15.1.1. High Frequency Oscillatory Ventilator (HFOV or Oscillator) has active exhalation and increased expiratory gas velocity compared to HFJV. Consider it for:

5.15.1.1.1. Severe Respiratory Distress Syndrome (RDS)

5.15.1.1.2. Acute Respiratory Distress Syndrome (ARDS)

5.15.1.2. Before initiating HFV review:

5.15.1.2.1. Pathophysiology of the disease process

5.15.1.2.2. Patient suitability

5.15.1.2.3. Patient diagnosis

5.15.1.2.4. Adequate fluid volume status

5.15.1.2.5. Adequate cardiac contractility (heart must maintain adequate performance with constant MAP from oscillation)

5.15.1.3. Initiate HFV when there is inability to adequately oxygenate or ventilate using the conventional ventilation to its limits. (For example tidal volume required greater than 5ml/kg, rate set at 60 bpm with the pCO2 at an unacceptably high level.) Consider HFV when the following circumstances persist for at least 2 hours after a dose of artificial surfactant:

5.15.1.3.1. The MAP on conventional ventilation is greater than 12.0 cm H2O.

5.15.1.3.2. The tidal volume set on the ventilator of 5ml/kg is not adequate to ventilate the patient. Tidal volume is set to birth weight for the first 2 weeks of life. Dry weight should be used if the baby is edematous.

5.15.1.3.3. The peak inspiratory pressure (PIP) is consistently greater than 25cm H2O, despite all efforts to optimize conventional ventilation.

5.15.1.3.4. The FiO2 cannot be weaned to less than 60% despite all efforts to optimize conventional ventilation.

5.15.1.4. Relative contraindications for HFOV:

5.15.1.4.1. Poor cardiac function

5.15.1.4.2. Dehydration state (low fluid volume)

5.15.2. **Management Guidelines**

5.15.2.1. Monitor the patient with a transcutaneous oxygen saturation monitor.

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5.15.2.2. Do chest X-rays as often as required to help determine lung inflation.

5.15.2.3. Do a blood gas within 30 minutes of initiating HFV and after any setting change.

5.15.2.4. Have an RT present during endotracheal suctioning, depending on the clinical situation, in some cases it may be necessary immediately prior to suctioning the RT to decrease iNO to zero, suction, then immediately increase the iNO back to preset value in order to avoid large increase in iNO.

5.15.3. **HFOV**

5.15.3.1. **MAP**: Start 2 cm H2O Higher than the MAP on conventional ventilation. Increased until the lung is open (demonstrated by chest x-ray), and O2 saturation is acceptable and the FiO2 can be weaned. Once MAP has been optimized consider weaning the MAP.

5.15.3.2. **Amplitude** (Delta Pressure) used for CO2 management:

5.15.3.2.1. The baby must have adequate chest movement described as 'gently wiggling down to the abdomen'. Consider starting with amplitude double the MAP then adjust according to blood gas and X-rays.

5.15.3.2.2. If the delta pressure (Amplitude) is close to 3 times greater than the MAP, consider decreasing the frequency (i.e. avoid getting close to delta P: MAP ratio of 3:1).

5.15.3.3. **Frequency**: Set at 8 Hz and adjust to the clinical pathology of the patient.

5.15.3.3.1. Consider lower frequency when air trapping is suspected.

5.15.3.3.2. If frequency is decreased, consider decreasing delta pressure.

5.15.3.3.3. Observe chest wiggle before and after frequency change.

6.0 EQUIPMENTS

6.1. Mechanical Ventilator

7.0 ATTACHMENT/FORMS

7.1. Nurse Observation Form (Form No 43)
7.2. Nurses Progress Notes (GDOH-NUR-NPN-203)
7.3. Vital signs Chart (GDOH-NUR-VSC-193)
7.4. Newborn Nursing Initial/Admission Assessment Form (GDOH-NUR-NNIAA-213)
7.5. Newborn Nursing Daily Re-Assessment Form (GDOH-NUR-NNDR-219)

8.0 REFERENCES

8.1. Neonatal clinical practice guideline (2015), Winnipeg regional health Authority

8.2. American Academy of Pediatrics, Fetus and Newborn Committee (2014) Respiratory support in preterm infants at birth (2014). Pediatrics, 133(1), 171-174.

8.3. www.iosjournals.org: Caring Newborn on Ventilator

9.0 REVISION CLAUSE

9.1. Modification of policy format, lay out and policy coding

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General Directorate of Health Affairs
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وزارة الصحة
المديرية العامة للشئون الصحية بمنطقة الباحة

HOSPITAL :

Health Care No .: ☐☐☐☐☐☐☐☐☐☐☐☐

Name : .....

Age : ☐☐ Yrs./Month Sex : ☐M ☐F

Nationality : .....

Consultant In-Charge : .....

Dept : ..... Unit : .....

NURSES OBSERVATION FORM نمـوذج مـلاحظـات التمـريضـ

Nurse / Date	Time	Temp.	Puls Rate	Resp Rate	B.P.	Pupils		Level of Consciousness Muscle Tone &Reflex	Cough Reflex	Response to Stimuli		Perception Pulses			Skin Color	Rema-rks	Sign.
						R	L			Verbal	Painful	ORAL	Audio Aural	visual			

SPECIFIC NURSING CARE COMMENTS :

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Ministry of Health

Hospital: _____ مستشفى:

Region: _____ المنطقة/المحافظة:

Dept./Unit: _____ القسم/الوحدة:

MRN: | | | | | | | | | | رقم الملف الطبي:

Name: _____ الاسم:

Nationality: _____ الجنسية:

Age: _____ ☐ سنة Years ☐ شهر Months ☐ يوم Days العمر:

Date of Birth: ___ / ___ / 14___ H ___ / ___ / 20___ تاريخ الميلاد:

Gender: ☐ Male ☐ Female الجنس:

NURSES PROGRESS NOTES FORM

DATE &TIME	NURSES NOTES (D-data A-action R-response)	REGISTERED NURSE INITIAL WITH ID NUMBER

Note: Write the time in each entry & affix your initial at the end of each paragraph. Document your complete Name, Initial, Job number, Date & Time at the closure of your documentation. Draw a line across empty spaces.

0 000000 002035 SN

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Dept./Unit: _____ القسم/الوحدة:

MRN: _____ رقم الملف الطبي:

Name: _____ الاسم:

Nationality: _____ الجنسية:

Age: _____ ☐ سنة Years ☐ شهر Months ☐ يوم Days العمر:

Date of Birth: ___/___/14___ H ___/___/20___ تاريخ الميلاد:

Gender: ☐ Male ☐ Female الجنس:

VITAL SIGNS CHART

WEIGHT-kg						
POST-OPERATIVE DAY						
DATE						
TIME						
TEMPERATURE	40					
	39					
	38					
	37					
	36					
	35					
RESPIRATORY RATE						
O_2 SATURATION						
BLOOD PRESSURE AND PULSE RATE PULSE = X SYSTOLIC = V DIASTOLIC = ^	240					
	220					
	200					
	180					
	160					
	140					
	120					
	100					
	80					
	60					
	40					
	20					
BP						
PAIN SCORE SCALE						
INITIALS AND JOB NUMBER						

0 000000 001939

SN _____

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MRN: | | | | | | | | | | | رقم الملف الطبي:

Name: _____ الاسم:

Nationality: _____ الجنسية:

Age: _____ ☐ سنة Years ☐ شهر Months ☐ يوم Days العمر:

Date of Birth: ____ / ____ / 14____ H ____ / ____ / 20____ تاريخ الميلاد:

Gender: ☐ Male ☐ Female الجنس:

NEWBORN NURSING INITIAL/ ADMISSION ASSESSMENT FORM

I. ADMISSION SOURCE:

☐ ER ☐ LR ☐ OR ☐ OTHERS.....

II. ADMISSION DIAGNOSIS

.....

III. BIRTH HISTORY:

BIRTH DATE: / TIME:	V. ADMISSION DATE/ TIME:	APGAR: 1minute	APGAR: 1minute	ID BRACELET #:

TYPE OF DELIVERY:
VAGINAL..... C-SECTION .....
OTHERS .....

WEEKS OF GESTATION: .....

BLOOD TYPE / Rh .....

PARAMETERS:
BIRTHWEIGHT..... gm (kg) / LENGTH .....cm / HEAD CIRCUMFERENCE .....cm / CHEST CIRCUMFERENCE.....cm

RESUSCITATION:
NONE ..... OXYGEN..... BAG / MASK ..... INTUBATION ..... CPR .....

HEP «B» VACC:
SITE..... DATE:.....
TIME:.....

VIT K:
SITE ..... TIME
..... INT.....

ERYTHROMYCIN (PROPHYLACTIC EYE TREATMENT)
DATE/TIME:..... INT.....

IV. MATERNAL HISTORY:

MATERNAL AGE:	MAT BLOOD TYPE:	G /Para:	GBS:	PROM ..... HRS	MAT TEMP:	OTHER:

III. PHYSICIAN NOTIFIED
☐ YES ☐ NO

PHYSICIAN NAME:

TIME:

V. TRANSITION NOTE:

DATE/TIME	TEMP	HR	RR	SPO2	BP	BLOOD GLUCOSE	ACTIVITY	COLOR	URINE	STOOL	OTHER	INITIAL

ACTIVITY OBSERVATIONS ++= SPONTANEOUS +=WITH STIMULATION L=LMP

VI. PHYSICAL ASSESSMENT:

CATEGORY	OBSERVATIONS	COMMENTS (Finding Indicated by * require notes)
General Appearance	COLOR: ☐pink ☐pale ☐acrocyanosis* ☐jaundice*	
	CRY: ☐strong ☐weak ☐high-pitched*	
	TONE: ☐good tone ☐hypotonic* ☐hypertonic*	
	MATURITY: ☐term ☐pre-term ☐post-term	
Skin	☐Clear ☐Peeling ☐Rash* ☐Bruising*	
	☐Vernix ☐Petechiae * ☐Mongolian spot	
Head	☐intact ☐Molding ☐Caput ☐Bruising*	
	☐Open Flat Fontanels ☐Cephalohematoma	
Eyes	☐Clear ☐Discharge* ☐Jaundice* ☐Hemorrhage	
ENT	☐Intact ☐Palate ☐Normal Ear Setting	
	☐Patent Nares ☐Nasal Flaring	

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Name:_____ الاسم:	MRN: _____ رقم الملف الطبي:

Thorax	❑Symmetrical ❑Clavicle (intact / fractured)	
Lungs	❑Clear ❑Equal Expansion Bilaterally ❑Retractions*	
	❑Grunting* ❑Coarse ❑Breath Sounds*Abd. .....cms ❑Abdomen Soft / Distended*	
Heart	❑Regular Rate ❑Peripheral Pulses Bilaterally (Y/N)	
Abdomen	Abd.....cms Abdomen ❑Soft / Distended*	
	Umbilical cord ❑Bowel Sounds (present / diminished*/ absent*)	
Genitalia	❑Male ❑Female ❑Ambiguous*	
	Testes: L ..... R..... Discharge*	
Anus	Patent ❑Meconium (present* / absent*)	
Trunk – Spine	Gluteal Folds (equal/unequal*) ❑Hip Click (R/L)	
Extremities	Symmetrical ❑Extra Digits* ❑Syndactyly	
Reflexes (noted)	Moro ❑ Grasp ❑Suck ❑Swallow	

VII. NUTRITIONAL ASSESSMENT:

FIRST FEED: ❑Breast Feed ❑Bottle Feed: Time ..... Type ..... Amount.....

VIII.PAIN ASSESSMENT: CRIES NEONATAL PAIN SCALE

- Any score above 4 indicates pain and infant should receive pain management intervention.

CATEGORY	PARAMETERS	SCORE	PATIENT'S SCORE
Crying	No	0	
	High Pitched (Consolable)	1	
	Inconsolable	2	
Requires 0_2 for Sat greater than 95%	No	0	
	Less than 30%	1	
	greater than 30%	2	
Increased vital signs	HR, BP within 10% of Pre-Op value	0	
	11% to 20% greater than Pre-Op values	1	
	greater than 21% of Pre-Op values	2	
Expression	None	0	
	Grimace	1	
	Grimace/Grunt	2	
Sleeplessness	No	0	
	Wakes at frequent intervals	1	
	Constantly awake	2	
	TOTAL PATIENT'S PAIN SCORE		

SCORING: ❑0- 3 No pain ❑4- 6 Moderate pain ❑7- 10 Severe pain

Note: Grimace consists of lowered brow, eyes squeezed shut, deepening nasolabial furrow, and open eyelids.
Non-audible grunt - only heard with a stethoscope

IX."HUMPTY DUMPTY"FALL RISK ASSESSMENT: (Write & sum up the appropriate answer from "a" to 'g' to get the total)
SCORE: ..... (If score is 12 or above at risk for falls) Minimum Score = 7 Maximum Score = 23

Parameters	Criteria	Score	Patient's Score
a) Age	Less than 3 years old	4	
	3 to less than 7 years old	3	
	7 to less than 13 years old	2	
	13 years and above	1	
b) Gender	Male	2	
	Female	1	
c) Diagnosis	Neurological Diagnosis	4	
	Alterations in Oxygenation (Respiratory Diagnosis, Dehydration, Anemia, Anorexia, Syncope/Dizziness, etc.)	3	
	Psychological/Behavioral Disorders	2	
	Other Diagnosis	1	
d) Cognitive Impairments	Not aware of Limitation	3	
	Forgets Limitations	2	
	Oriented to own ability	1	

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e)Environmental Factors	History of falls or Infant-Toddler placed in bed	4	
	Patient uses assistive devices or Infant-Toddler in crib or Furniture/Lighting (Tripled room)	3	
	Patient placed in bed	2	
	Outpatient Area	1	
f) Response to Surgery/Sedation/ Anesthesia	Within 24 hours	3	
	Within 48 hours	2	
	More than 48 hours/None	1	
g) Medication Usage	Multiple usage of : Sedatives (excluding ICU patients sedated and paralyzed) Hypnotics, Barbiturates, Phenothiazines, Antidepressants, Laxatives/Diuretics, Narcotics	3	
	One of the medication listed above	2	
	Other medications/None	1	
		TOTAL PATIENT›S FALL RISK SCORE	

X. SKIN RISK ASSESSMENT: NEONATAL / INFANT BRADEN Q SCALE (NIBQS)

(Write number adjacent to descriptor; add for total score)

Parameters	Criteria	Score	Patient's Score
a) Gestational age	Less than 28 wks.	1	
	greater than 28- Less than 33 wks.	2	
	greater than 33- Less than 38 wks.	3	
	greater than 38wks.	4	
b) Mobility	Completely immobile	1	
	Very limited	2	
	Slightly limited	3	
	No limitation	4	
c) Activity	Bedfast	1	
	Very limited	2	
	Slightly limited	3	
	No limitation	4	
d) Sensory perception	Completely limited	1	
	Very limited	2	
	Slightly limited	3	
	No impairment	4	
e) Moisture	Constantly moist	1	
	Very moist	2	
	Occasionally moist	3	
	Rarely moist	4	
f) Friction/ Shear	Significant problem	1	
	Problem	2	
	Potential problem	3	
	No apparent problem	4	
g) Nutrition	Very poor	1	
	Inadequate	2	
	Adequate	3	
	Excellent	4	
h) Tissue perfusion & oxygenation	Extremely compromised	1	
	Compromised	2	
	Adequate	3	
	Excellent	4	
		TOTAL PATIENT ‹S SKIN RISK SCORE	

Score:..... If Less than 20At risk for skin breakdown

Diaper Dermatitis risk:* (identification of one or more risk factors+ enteral feeding = dermatitis risk.)

❑Frequent stool ❑ Bowel surgery ❑ Short gut ❑hyper-caloric feeding ❑PGEs ❑On Antibiotics ❑ Prolonged NPO status

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Name:_____ الاسم: | MRN: |_|_|_|_|_|_|_|_|_|_| رقم الملف الطبي:

Interventions: ❑Skin cleansing/protection ❑Gel pillow ❑Sheepskin ❑Scheduled turning ❑Tegaderm
❑Reduce friction /shear ❑Petroleum jelly ointment ❑Desitin ointment ❑Citric acid ointment ❑Other:.....
Comments:.....

XI.OTHERS: (oral and nasogastric tubes, dressing, restraint (splint), umbilical catheter)

.....

XII.SAFETY:

❑Cardio-respiratory audible alarms at 70% volume ❑Oximeter alarm settings: Low..... High.....
❑Bag/mask/suction@ bedside: FiO2..... ❑ IV fluids/rate verified ❑High risk medication infusion dose/rate verified
❑Bed appropriate for developmental level
❑Radiant warmer ❑Incubator; NTE..... ❑Bassinette ❑NICU Crib ❑Pedi Crib
❑I.D Band x 2 ❑I.D. band location:1..... 2..... MR#.....

XIII. PSYCHOSOCIAL :

Patient / family express or demonstrate coping: ❑Yes ❑No
Family active in care: ❑Yes ❑No Detail:.....
Support needs identified:
❑Emotional support ❑Interpreter ❑Social worker ❑Chaplain ❑Lactation consultant

XIV. DISCHARGE PLANNING

SOCIOECONOMIC NEEDS:		
Lack of needed caregiver; family support	❑Yes	❑No
At risk of abuse or neglect	❑Yes	❑No
Inadequate resources: insurance, financial	❑Yes	❑ No
Foster parent, guardian etc.	❑Yes	❑No
PHYSICAL NEEDS:		
Metabolic Screening	❑Yes	❑No
ENVIRONMENTAL NEEDS:		
Change in living arrangements	❑Yes	❑No
In-home care or equipment	❑Yes	❑No
REFERRAL INDICATED:		
Referral sent to:		
Social Services	❑Yes	❑No
Home Care	❑Yes	❑No
	❑Yes	❑No
Others.....	❑Yes	❑No
High risk indicated but no referral sent, why?		

XIII. EDUCATIONAL / GENERAL NEEDS:

Repeated, unscheduled admissions	❑Yes	❑No
Newly diagnosed chronic/ terminal illness	❑Yes	❑No
Family education needed for in-home care	❑Yes	❑No
Immunization awareness	❑Yes	❑No

PHYSICAL DEFICITS

Cardiovascular	❑Yes	❑No
example		
Respiratory	❑Yes	❑No
example		
Neurological	❑Yes	❑No
example		
Sensory/Speech	❑Yes	❑No
example		
Gastrointestinal/Nutritional	❑Yes	❑No
example		
Genitourinary	❑Yes	❑No
example		
Musculoskeletal/Mobility	❑Yes	❑No
example		
Skin/Wound	❑Yes	❑No
example		
Cognitive/Mental	❑Yes	❑No
example		
Endocrine	❑Yes	❑No
example		
Language Barrier	❑Yes	❑No
example		
Other Concerns:		

....

RN NAME (Assessor) **Signature** **Designation** **Job number** **Date & Time**

Note: Please fill-up the data required completely and legibly. Put check ( ) if applicable. Use military time on your documentation. Affix your initial, date and time at the bottom of each page. Draw a line across empty spaces.

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Name:_____ الاسم:	MRN: رقم الملف الطبي:

XV. PROBLEM SUMMARY LIST
"ACTION : REFER TO MULTIDISCIPLINARY PLAN OF CARE FORM"

SERIAL NUMBER	ASSESSMENT / PATIENT NEEDS FINDINGS

RN NAME | Signature | Designation

Job number | Date (dd/mm/yy) | Time

NOTE: Please fill-up the data required completely and legibly. Put check (√) if applicable. Use military time on your documentation. Affix your initial, date and time at the bottom of each page. Draw a line across empty spaces.

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SN

KINGDOM OF SAUDI ARABIA

وزارة الصحة
Ministry of Health

Hospital: _____ مستشفى:

Region: _____ المنطقة/المحافظة:

Dept./Unit: _____ القسم/الوحدة:

MRN: | | | | | | | | | | رقم الملف الطبي:

Name: _____ الاسم:

Nationality: _____ الجنسية:

Age: _____ ☐ سنة Years ☐ شهر Months ☐ يوم Days العمر:

Date of Birth: ____ / ____ / 14____ H ____ / ____ / 20____ تاريخ الميلاد:

Gender: ☐ Male ☐ Female الجنس:

NEWBORN NURSING DAILY RE-ASSESSMENT FORM

ALLERGY: ☐ No Known Allergies ☐ If yes, include allergy to medications, food & blood products ..... WEIGHT: ..... HEIGHT: .....

Date/Time ..... MORNING	Date/Time ..... EVENING	Date/Time ..... NIGHT
Received patient : I. Level of consciousness ☐ Alert ☐ Lethargic ☐ Stuporous ☐ Coma II. Position : _____ III. Airway : ☐ Maintains own ☐ Oral airway size _____ ☐ Nasal airway size _____ ☐ ETT size ____ ☐ TT size _____ ☐ Mechanical Ventilator : ☐ SIMV ☐ HFOV (☐ CMV ☐ Spontaneous FiO2____ O2____ Ventelatory rate____ PEEP____ TV____ ☐ Oxygen at ____ liter per minute ☐ Mask ☐ Nasal cannula IV. Breathing : a. Rhythm ☐regular ☐irregular ☐paradoxical b. Depth ☐normal ☐shallow ☐deep c. Quality ☐normal (☐labored ☐stridor d. Cough ☐none ☐productive ( )non-productive e. Bronchial/ lung sound ☐normal ☐wheeze ☐crackles ☐ ronchi V. Circulation: a. Pulse ☐ regular ☐ irregular ☐ normal ☐weak ☐bounding b. Skin ☐normal ☐pale ☐flushed	Received patient : I. Level of consciousness ☐ Alert ☐ Lethargic ☐ Stuporous ☐ Coma II. Position : _____ III. Airway : ☐ Maintains own ☐ Oral airway size _____ ☐ Nasal airway size _____ ☐ ETT size ____ ☐ TT size _____ ☐ Mechanical Ventilator : ☐ SIMV ☐ HFOV (☐ CMV ☐ Spontaneous FiO2____ O2____ Ventelatory rate____ PEEP____ TV____ ☐ Oxygen at ____ liter per minute ☐ Mask ☐ Nasal cannula IV. Breathing : a. Rhythm ☐regular ☐irregular ☐paradoxical b. Depth ☐normal ☐shallow ☐deep c. Quality ☐normal (☐labored ☐stridor d. Cough ☐none ☐productive ( )non-productive e. Bronchial/ lung sound ☐normal ☐wheeze ☐crackles ☐ ronchi V. Circulation: a. Pulse ☐ regular ☐ irregular ☐ normal ☐weak ☐bounding b. Skin ☐normal ☐pale ☐flushed	Received patient : I. Level of consciousness ☐ Alert ☐ Lethargic ☐ Stuporous ☐ Coma II. Position : _____ III. Airway : ☐ Maintains own ☐ Oral airway size _____ ☐ Nasal airway size _____ ☐ ETT size ____ ☐ TT size _____ ☐ Mechanical Ventilator : ☐ SIMV ☐ HFOV (☐ CMV ☐ Spontaneous FiO2____ O2____ Ventelatory rate____ PEEP____ TV____ ☐ Oxygen at ____ liter per minute ☐ Mask ☐ Nasal cannula IV. Breathing : a. Rhythm ☐regular ☐irregular ☐paradoxical b. Depth ☐normal ☐shallow ☐deep c. Quality ☐normal (☐labored ☐stridor d. Cough ☐none ☐productive ( )non-productive e. Bronchial/ lung sound ☐normal ☐wheeze ☐crackles ☐ ronchi V. Circulation: a. Pulse ☐ regular ☐ irregular ☐ normal ☐weak ☐bounding b. Skin ☐normal ☐pale ☐flushed

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Name:_____ الاسم: | MRN: |_|_|_|_|_|_|_|_|_|_| رقم الملف الطبي:

❑ warm ❑ cyanotic ❑ jaundice
❑ mottled ❑cool ❑ dry
❑ diaphoretic ❑ edema
c. IV Line ❑ Yes ❑ No
Site_____
d. IV Fluids _____ at
_____ ml level _____ drops ml/hrVI.
Nutrition:
a) Feeding
❑ Breast feed ❑ Bottle feed
b) Tolerance
❑ good ❑ fair ❑ poor
❑ nausea
❑ vomiting ❑ gastric aspiration
c) ❑ NGT size _____
d) ❑ OGT size _____
VII. Elimination:
a) Bowel movement
❑ Normal ❑ Constipation
❑ Loose ❑ Colostomy
b) Urine
❑ Normal ❑Polyuria ❑ Oliguria
❑ Amber ❑Cloudy ❑Hematuria
❑ Urine collection bag
Amount _____ml.
VIII. Pain and Discomfort Present:
❑Yes ❑ No
IX. Others:_____
RN NAME:_____
Initial: _____Job Number_____
VI. Nutrition
a. Diet _____
b. Appetite
❑ good ❑ fair ❑ poor
❑ nausea ❑ vomiting
c.( ) NGT size ____ d. ❑ GT size ____
VII. Elimination
a. Bowel movement
❑normal ❑constipation
❑loose ❑ colostomy
b. Urine:
❑normal ❑polyuria ❑oliguria
❑amber ❑cloudy ❑hematuria
❑ foley catheter F_____
❑ condom cath
Urine collection bag,
amount_____cc
VIII. Others:_____
RN NAME:_____
Initial: _____

❑ warm ❑ cyanotic ❑ jaundice
❑ mottled ❑cool ❑ dry
❑ diaphoretic ❑ edema
c. IV Line ❑ Yes ❑ No
Site_____
d. IV Fluids _____ at
_____ ml level _____ drops ml/hrVI.
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❑ nausea
❑ vomiting ❑ gastric aspiration
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❑ Normal ❑Polyuria ❑ Oliguria
❑ Amber ❑Cloudy ❑Hematuria
❑ Urine collection bag
Amount _____ml.
VIII. Pain and Discomfort Present:
❑Yes ❑ No
IX. Others:_____
RN NAME:_____
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b. Urine:
❑normal ❑polyuria ❑oliguria
❑amber ❑cloudy ❑hematuria
❑ foley catheter F_____
❑ condom cath
Urine collection bag,
amount_____cc
VIII. Others:_____
RN NAME:_____
Initial: _____

❑ warm ❑ cyanotic ❑ jaundice
❑ mottled ❑cool ❑ dry
❑ diaphoretic ❑ edema
c. IV Line ❑ Yes ❑ No
Site_____
d. IV Fluids _____ at
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❑ Loose ❑ Colostomy
b) Urine
❑ Normal ❑Polyuria ❑ Oliguria
❑ Amber ❑Cloudy ❑Hematuria
❑ Urine collection bag
Amount _____ml.
VIII. Pain and Discomfort Present:
❑Yes ❑ No
IX. Others:_____
RN NAME:_____
Initial: _____Job Number_____
VI. Nutrition
a. Diet _____
b. Appetite
❑ good ❑ fair ❑ poor
❑ nausea ❑ vomiting
c.( ) NGT size ____ d. ❑ GT size ____
VII. Elimination
a. Bowel movement
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b. Urine:
❑normal ❑polyuria ❑oliguria
❑amber ❑cloudy ❑hematuria
❑ foley catheter F_____
❑ condom cath
Urine collection bag,
amount_____cc
VIII. Others:_____
RN NAME:_____
Initial: _____

Note: Please fill up the data required (Nursing Daily Re- Assessment) completely and legibly during endorsement of the outgoing to incoming shift and upon receipt of the trans- in patient. Put check (√) if applicable. Use military time in your documentation. Draw a line across empty spaces.

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Name:_____ الاسم:	MRN: رقم الملف الطبي:

XI. NEONATAL / INFANT "BRADEN Q SCALE (NIBQS) " SKIN RISK RE-ASSESSMENT
(Write number adjacent to descriptor and sum up for total score)

CATEGORIES	PARAMETERS	Skin Risk Score			
		TIME			
a) Gestational age	Greater than 38 Weeks	4			
	Greater than 33-less than or equal to 38 Weeks	3			
	Greater than 28- less than or equal to 33 Weeks	2			
	Less than28- less than or equal to 33 Weeks	1			
b) Mobility	No limitation	4			
	Slightly limited	3			
	Very limited	2			
	Completely Immobile	1			
c) Activity	No limitation	4			
	Slightly limited	3			
	Very limited	2			
	Bedfast	1			
d) Sensory perception	No impairment	4			
	Slightly limited	3			
	Very limited	2			
	Completely limited	1			
e) Moisture	Rarely moist	4			
	Occasionally moist	3			
	Very moist	2			
	Constantly moist	1			
f) Shear and friction	No apparent problem	4			
	Potential problem	3			
	Problem	2			
	Significant problem	1			
g) Nutrition	Excellent	4			
	Adequate	3			
	Inadequate	2			
	Very poor	1			
h) Tissue perfusion and Oxygenation	Excellent	4			
	Adequate	3			
	Compromised	2			
	Extremely compromised	1			
		PATIENT'S SCORE			
		INITIAL & JOB NO.			

Score of less than 20At risk for skin breakdown

Note: Perform "NIBQS" Skin risk re- assessment if on assessment is indicated and if potential problems occur during patient's hospitalization. Draw a line across empty spaces.

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